

WRANGELL, AK, USA

WMO: 703870

Lat: 56.483N

Lon: 132.367W

Elev: 13

StdP: 101.17

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25338

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-8.7	-19.0	0.7	-7.6	-17.3	0.8	-7.0	11.5	3.4	10.9	2.0	4.0	80	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.4	22.0	15.6	19.8	14.5	17.9	13.4	15.9	20.5	15.0	18.5	14.2	17.3	2.9	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.8	9.9	17.1	13.0	9.3	16.1	12.6	9.1	15.6	44.5	20.8	41.7	18.5	39.8	17.2	19.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.7	7.7	-12.5	25.3	2.2	1.8	-14.0	26.6	-15.3	27.7	-16.6	28.7	-18.2	30.0		
(5)			-13.7	17.6	1.9	0.8	-15.0	18.2	-16.2	18.7	-17.3	19.2	-18.6	19.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	0.6	1.3	3.0	6.1	9.9	12.7	14.0	13.9	11.3	7.3	3.5	1.1
	DBStd	5.77	4.82	3.75	3.01	2.20	2.58	2.34	1.80	1.87	1.81	2.30	3.18	4.12
	HDD10.0	1471	292	244	218	118	33	2	0	0	7	86	195	276
	HDD18.3	4104	550	478	476	366	260	171	135	138	212	341	445	534
	CDD10.0	410	0	0	0	2	31	82	124	121	46	3	0	0
	CDD18.3	2	0	0	0	0	0	1	1	1	0	0	0	0
	CDH23.3	16	0	0	0	0	0	8	3	4	0	0	0	0
	CDH26.7	1	0	0	0	0	0	1	0	0	0	0	0	0
(14) Wind	WSAvg	2.9	3.6	3.3	3.5	3.1	2.6	2.2	1.8	1.9	2.4	3.2	3.6	3.8
(15) Precipitation	PrecAvg	2010	185	138	129	120	112	106	116	143	225	332	214	190
	PrecMax	2430	370	250	246	211	189	171	227	327	432	595	445	324
	PrecMin	1633	41	0	38	20	44	38	35	35	44	195	42	48
	PrecStd	219	81	68	50	48	40	32	47	68	90	102	105	71
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	9.0	12.1	17.3	21.3	24.1	23.7	22.9	18.8	14.1	11.2	9.7
		MCWB	6.6	5.6	7.1	9.8	12.8	16.2	16.8	16.4	13.8	10.4	8.2	6.7
	2%	DB	7.7	7.7	9.0	13.5	18.6	21.2	21.3	21.1	16.6	12.5	9.1	8.0
		MCWB	5.7	5.0	5.7	7.8	11.8	14.7	15.8	15.4	12.5	10.1	7.5	6.0
	5%	DB	7.2	7.1	7.7	11.5	16.8	18.7	18.9	18.9	15.1	11.3	8.0	7.3
		MCWB	5.1	4.6	4.9	6.7	10.8	13.6	14.6	14.7	12.0	9.3	6.3	5.4
	10%	DB	6.1	6.1	7.0	9.8	14.1	17.1	17.4	17.3	13.9	10.4	7.4	6.2
		MCWB	4.2	4.0	4.4	6.2	9.6	12.6	13.7	13.8	11.4	8.5	5.6	4.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	6.3	7.6	10.2	13.9	16.7	17.2	16.9	14.9	11.8	9.0	7.8
		MCDB	8.4	8.1	10.3	15.9	19.8	23.5	22.7	22.3	17.4	13.1	10.2	8.8
	2%	WB	6.1	5.4	6.3	8.3	12.4	15.0	16.0	15.8	13.6	10.5	7.9	6.6
		MCDB	7.6	7.3	8.6	12.5	17.4	19.9	20.7	19.8	15.6	12.0	9.0	8.1
	5%	WB	5.2	4.8	5.3	7.4	11.3	13.9	15.0	15.0	12.7	9.5	6.8	5.3
		MCDB	7.0	6.8	7.5	10.3	15.3	17.9	18.2	18.1	14.5	11.0	8.1	7.1
	10%	WB	4.5	4.2	4.6	6.6	10.4	12.9	14.1	14.2	11.9	8.6	5.7	4.5
		MCDB	6.2	6.1	6.7	9.3	13.7	16.1	16.9	16.9	13.3	9.9	7.3	6.2
(35) Mean Daily Temperature Range	5% DB	MDBR	3.6	3.9	4.7	6.0	6.7	6.1	5.4	5.6	4.8	3.8	3.4	3.6
		MCDBR	4.2	4.6	6.2	9.7	10.9	10.6	9.7	9.7	7.2	5.1	4.2	4.2
	5% WB	MCWBR	3.1	3.1	4.0	5.3	5.7	5.7	5.3	5.4	4.2	3.3	3.0	3.1
		MCDWR	4.1	4.2	5.5	8.0	9.2	9.0	8.2	8.2	5.9	4.3	4.1	4.0
(40) Clear-Sky Solar Irradiance	taub		0.286	0.302	0.331	0.376	0.396	0.385	0.402	0.369	0.356	0.328	0.291	0.276
		taud	2.211	2.298	2.240	2.143	2.169	2.322	2.326	2.451	2.455	2.413	2.229	2.137
	Ebn at Noon		598	759	823	834	840	857	832	841	798	722	581	492
		Edh at Noon	54	74	101	130	135	118	115	96	83	67	52	44
(44) All-Sky Solar Radiation	RadAvg		0.45	1.22	2.25	3.69	4.72	4.75	4.13	3.54	2.29	1.25	0.56	0.34
	RadStd		0.06	0.24	0.34	0.36	0.54	0.42	0.55	0.41	0.37	0.19	0.09	0.06

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.43	+0.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+66	N/A

Nomenclature: See separate page